Further Strategies — problem 14.42

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(a) Prove that there exist integers a, b, c not all zero and each of absolute value less than one million, such that $|a + b\sqrt{2} + c\sqrt{3}| < 10^{-11}$.

(b) Let a, b, c be integers, not all zero and each of absolute value less than one million. Prove that $|a + b\sqrt{2} + c\sqrt{3}| > 10^{-21}$ (Putnam 1980).